THE HONG KONG UNIVERSITY OF SCIENCE AND TECHNOLOGY | Undergraduate Program & Course Catalog

Undergraduate Courses 2026-27

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FINA | Finance

FINA 1303 Introduction to Financial Markets and Institutions 3 Credit(s)

Exclusion(s) *FINA 3103, CORE 1330*

Description

A broad introduction to financial institutions and financial markets, providing students with an understanding of how financial systems work, the functions of various financial institutions and the role of financial regulation.

Intended Learning Outcomes

On successful completion of the course, students will be able to:

1. Explain and apply basic interest rate concepts in particular Discounted Cash Flow (DCF)
2. Calculate interest on basic financial instruments
3. Apply simple pricing theories such as the Dividend Discount Model (DDM)
4. Explain the function and organization of financial markets and institutions and the interactions between them
5. Explain ethical and professional responsibility of finance professionals through the discussion of current social responsibility and ethical issues in finance during class

FINA 2203 Fundamentals of Business Finance 3 Credit(s)

Exclusion(s) *FINA 2303*

Description

For BIBU and non-SB&M students. This course provides an introduction to corporate finance for students who may not have background in accounting and business statistics. Topics include financial statement analysis, time value of money, bond and stock valuation, capital budgeting, risk and return concepts and cost of capital.

Intended Learning Outcomes

On successful completion of the course, students will be able to:

1. Analyze firm financial performance through basic financial statement analysis.
2. Apply the concept of time value of money to make correct financial decisions.
3. Analyze the attractiveness of investment projects using various capital budgeting techniques, and understand the merits and drawbacks of each technique.
4. Explain the fundamental principles underlying asset valuation, and compute asset prices using a variety of approaches.
5. Explain the key concepts of risk, return, and diversification.
6. Explain the key concepts and implications of cost of capital.

– **FINA 2303** **Financial Management** **3 Credit(s)**

Prerequisite(s)	*ACCT 2010 AND (CIVL 2160 OR IEDA 2520 OR IEDA 2540 OR ISOM 2500 OR LIFS 3150 OR MATH 2411)*
Exclusion(s)	*FINA 2203*

Description

For SB&M students only. This course provides an introduction to corporate finance. Topics include time value of money, bond and stock valuation, capital budgeting, risk and return concepts and cost of capital.

Intended Learning Outcomes

On successful completion of the course, students will be able to:

1. Analyze firm financial performance through basic financial statement analysis.
2. Apply the concept of time value of money to make correct financial decisions.
3. Analyze the attractiveness of investment projects using various capital budgeting techniques, and understand the merits and drawbacks of each technique.
4. Explain the fundamental principles underlying asset valuation, and compute asset prices using a variety of approaches.
5. Explain the key concepts of risk, return, and diversification.
6. Explain the key concepts and implications of cost of capital.

– FINA 3001 Key Skills for Finance Professionals (A) 1 Credit(s)

Prerequisite(s) *FINA 2203 OR FINA 2303*

Description

The course aims to develop students' competitive edge so they are well prepared for their internships and future careers. Graded P, PP, or F. For students with major in FINA only. Students are required to complete the course within four regular terms.

Intended Learning Outcomes

On successful completion of the course, students will be able to:

1. Acquire in-depth knowledge and insights in current finance topics.
2. Acquire necessary skills to develop own professional identity.
3. Develop interpersonal, communication and organizational skills and build networking relationship with peers, practitioners and professionals.
4. Develop analytical skills, research skills, and presentation skills with actual hands-on experience.
5. Perform with professional and ethical code of conduct.

– FINA 3103 Intermediate Investments 3 Credit(s)

Prerequisite(s) *FINA 2203 or FINA 2303*

Description

This course studies the concepts and evidence relevant to the management of investment portfolios. Topics include diversification, asset allocation, portfolio optimization, factor models, the relation between risk and return, trading, passive (e.g., index-fund) and active (e.g., hedge-fund, long-short) strategies, mutual funds, performance evaluation, long-horizon investing and simulation.

Intended Learning Outcomes

On successful completion of the course, students will be able to:

1. Allocate capital across different asset classes based on established financial theories.
2. Evaluate and price financial assets using appropriate valuation models.
3. Assess portfolio performance to determine the effectiveness of portfolio management strategies.
4. Compare and contrast various investment strategies, including passive and active portfolio management.
5. Conduct financial analysis using quantitative tools and techniques.

FINA 3111 Finance Professional Development 1 Credit(s)

Description

The course aims to provide QFIN students with increased opportunities to acquire knowledge of the finance industry, connect with professionals and peers as well as to gain hands-on experience to develop interpersonal skills. The goal is to enhance their employability and long-term career success. Additionally, the course aims to promote peer support and strengthen community bonding through participation in department/program activities. Students are required to take the course for every regular term in which they are in residency at HKUST with major in QFIN. For students with major in QFIN only. Graded P/PP/F.

Intended Learning Outcomes

On successful completion of the course, students will be able to:

1. Acquire a better understanding of current finance topics.
2. Enhance interpersonal skills through connections with professionals and peers.
3. Apply financial knowledge through hands-on experiences.

FINA 3203 Derivative Securities 3 Credit(s)

Prerequisite(s)	*FINA 2203 OR FINA 2303*

Description

This course covers options, forward contracts, futures contracts and swaps. By the end of this course, students will have a good knowledge of how derivative contracts work, how they are used and how they are priced.

Intended Learning Outcomes

On successful completion of the course, students will be able to:

1. Understand the mechanics of the derivatives markets including futures, options, and swaps.
2. Use derivatives for hedging risks.
3. Use derivatives for trading views on direction and volatility.
4. Explain the principles of derivative pricing and analyze the impact of different factors on derivative prices.

FINA 3303 Intermediate Corporate Finance 3 Credit(s)

Prerequisite(s)	*FINA 2203 OR FINA 2303*
Cross-Campus Equivalent Course	*FTEC 3330*

Description

The course provides an in-depth treatment of long-term financing decisions, including estimation of the cost of capital, financial leverage, corporate distress, raising financing, dividend policy, and working capital analysis.

Intended Learning Outcomes

On successful completion of the course, students will be able to:

1. Apply finance theories to explain how firms choose their capital structure.
2. Value projects and firms using various valuation models, including discounted cash flow and relative valuation techniques.
3. Understand how firms raise capital and compute the cost of financing for various alternatives.
4. Understand payout policy, including dividends and share repurchases, and how factors such as taxes influence corporate decisions.
5. Understand how firms manage net working capital and analyze their effectiveness in doing so.

FINA 3810 Bloomberg Market Concepts Certification 0 Credit(s)

Prerequisite(s) *FINA 2203 OR FINA 2303*

Description

The course aims to provide students with knowledge of financial markets and Bloomberg functionality. The course consists of four modules: economic indicators, currencies, fixed income and equities. Students will need to complete all four modules to receive the Bloomberg Market Concepts certification. Graded P or F. For students with major in FINA, QFIN and SGFN only.

Intended Learning Outcomes

On successful completion of the course, students will be able to:

1. Use Bloomberg data, news, and tools to explain developments in global equity, fixed income, and currency markets.
2. Explain how key economic indicators are published and used in market analysis.
3. Navigate Bloomberg Terminal functionality to locate key financial data.

FINA 3820 Refinitiv Certification Program 0 Credit(s)

Prerequisite(s) *FINA 2303*

Description

This course will enable students to learn more about market information and the functionality of a Refinitiv data product most commonly used by financial professionals. It begins with the first module of basic overview and is followed by four core modules on data, screen design, graphics, analysis tools of various instruments, which includes one module of data retrieval in Excel. Students will be eligible in attending the Refinitiv Certification Assessment upon completion of the course. Graded P or F.

Intended Learning Outcomes

On successful completion of the course, students will be able to:

1. Use LSEG Workspace to access and interpret financial data, market-leading Reuters news, and analytics.
2. Apply LSEG Workspace tools and content to support informed financial decision-making.

FINA 3830 Bloomberg, LSEG with Artificial Intelligence Tools 0 Credit(s)

Prerequisite(s) *FINA 2203 OR FINA 2303*

Description

This course provides students with a hands-on, experiential introduction to professional financial data platforms and emerging AI technologies that are transforming the investment and analytics landscape. Students will develop practical skills using Bloomberg Terminal, London Stock Exchange Group (LSEG) Workspace, and a range of artificial intelligence tools commonly applied in investment analysis. Graded P or F.

Intended Learning Outcomes

On successful completion of the course, students will be able to:

1. Effectively navigate and retrieve financial market data using Bloomberg Terminal and LSEG Workspace across major asset classes and key macroeconomic indicators.
2. Apply AI enabled analytics tools to support financial modelling, data analysis, and investment research.
3. Integrate professional data platforms with human judgment to construct dashboards, research summaries, and coherent investment storylines.

FINA 4003 Wealth Management 3 Credit(s)

Prerequisite(s) *FINA 3103*

Description

This course focuses on a broad understanding of financial and investment issues arising in the context of personal wealth management. Students will gain an understanding of the nature and use of various financial products and legal instruments for developing a financial plan. It covers topics such as client financial status and expectations, portfolio management, securities trading, managed funds, insurance, and tax and estate planning.

Intended Learning Outcomes

On successful completion of the course, students will be able to:

1. Implement the financial planning process while following the professional practices.
2. Apply personal profiling and risk profiling techniques to comprehend the unique situation and preferences of the client.
3. Prioritize goals and determine the feasibility of achieving these goals.
4. Construct personal financial statements and use ratios and other techniques to analyze the client's financial situation.
5. Demonstrate the ability to develop and justify the economic assumptions used in a financial plan.
6. Construct optimal portfolios and make asset allocation decisions which match goals, risk tolerance and other client characteristics using Morningstar Direct.
7. Demonstrate the ability to analyze, synthesize, and evaluate personal financial situations, and implement goal funding strategies by determining realistic targets and conducting capital needs analysis for retirement and other financial objectives.
8. Demonstrate the ability to clearly discuss and justify recommendations.

FINA 4013 Corporate Valuation 3 Credit(s)

Prerequisite(s) *FINA 3303*

Description

This course teaches students how to use financial information to value firms. By the end of this course, students will be able to value firms using discounted cash flow valuation, relative valuation, and real options.

Intended Learning Outcomes

On successful completion of the course, students will be able to:

1. Analyze firm financial performance using financial statements to identify key value drivers.
2. Develop projected financial statements and forecast free cash flows for valuation purposes.
3. Estimate the cost of capital.
4. Apply discounted cash flow (DCF) models to value firms and conduct sensitivity analysis.
5. Use relative valuation techniques and other valuation methods to complement DCF analysis.
6. Apply valuation models to real world situations.

FINA 4103 Financial Markets Trading and Structure 3 Credit(s)

Prerequisite(s)	*FINA 3103*
Cross-Campus Equivalent Course	*FTEC 4310*

Description

This course examines liquidity, market structure and trading. The focus is on examining the efficiency of trading systems and markets, the impact of new algorithmic trading on markets, the role of intermediaries and regulatory actions on market structure issues. Students learn about a number of alternative market structures in terms of their economic and operational effectiveness.

Intended Learning Outcomes

On successful completion of the course, students will be able to:

1. Comprehend how to incorporate the notion of "information" in analyses of financial market.
2. Theoretically and empirically measure market liquidity.
3. Understand various financial market details and trading structures.
4. Understand the impact of recent development of financial technologies.
5. Analyze market quality in various situations with different market structures, trading rules, and market policies.

FINA 4203 Mergers, Acquisitions, and Corporate Restructuring 3 Credit(s)

Prerequisite(s) *FINA 3303*

Description

This course analyzes mechanisms underlying the creation of value in mergers, acquisitions and corporate restructuring. Students will learn analytical techniques and tools to analyze (a) strategy behind corporate restructuring transactions, whether and what to acquire, (b) evaluation and financing of acquisitions, and (c) decisions and processes that impact value creation in mergers.

Intended Learning Outcomes

On successful completion of the course, students will be able to:

1. Apply valuation techniques in M&A settings.
2. Analyze how M&A deals are financed and structured.
3. Evaluate the mechanisms and effectiveness of anti-takeover defenses.
4. Explain corporate restructurings, including divestitures and spinoffs.
5. Assess the role and impact of corporate governance in M&A transactions.

FINA 4303 Fixed Income Securities 3 Credit(s)

Prerequisite(s) *FINA 3103 AND FINA 3203*

Description

Pricing of bonds, term structure models, duration analysis and immunization techniques, valuation of interest rate options, convertible bonds, mortgage-backed securities, pricing credit derivatives, bond portfolio construction and management.

Intended Learning Outcomes

On successful completion of the course, students will be able to:

1. Evaluate and price a wide range of fixed income instruments—including traditional and structured bonds—using appropriate financial models and analytical tools.
2. Analyze interest rate risk using measures such as duration and convexity, and apply yield curve construction techniques including bootstrapping.
3. Design and assess fixed income portfolio management strategies, including immunization and hedging approaches, in the context of changing market conditions.
4. Understand and apply the mechanics and valuation of interest rate derivatives such as swaps, futures, and credit derivatives.
5. Evaluate emerging trends and innovations in fixed income markets, and assess their implications for investment strategies and risk management.

FINA 4403 International Finance 3 Credit(s)

Prerequisite(s) *FINA 3103*

Description

An introduction to the fundamental principles of international financial management and investment. Topics include: international financial markets and instruments; foreign exchange markets; foreign currency derivatives and currency risk; international capital budgeting; international bond and equity markets.

Intended Learning Outcomes

On successful completion of the course, students will be able to:

1. Demonstrate an understanding of the international dimensions of financial markets, including the international monetary system, capital markets, regulatory frameworks, risk factors, and investor behavior.
2. Apply finance concepts and theories to evaluate and make informed decisions in international business and investment contexts.
3. Apply economic principles to explain the determinants and dynamics of exchange rate movements.
4. Discuss how to use derivatives such as futures, options, and swaps to manage risks faced by multinational corporations.
5. Explain the principles of international portfolio management, including the relationship between risk and return across individual assets and investment portfolios.
6. Describe and explain recent developments related to financial markets outside Hong Kong.

– **FINA 4413** **Family Business** **3 Credit(s)**

Previous Course Code(s)	*FINA 4929N*
Prerequisite(s)	*FINA 2203 OR FINA 2303*

Description

This course aims to provide students with comprehensive and practical understanding of Asian family business, which is the most prominent and significant business form in Asia. The course will specifically cover valuation and ownership design, financing and exit strategies, governance and succession practices of family businesses. The course is application oriented and draws on the faculty's research experience and active interactions with family businesses in Asia. This course will benefit not only the students who will work in family businesses but also those likely to provide services to them in the future, such as private and investment bankers, family office professionals, private equity professionals, and other service professionals working closely with wealthy families in the region.

Intended Learning Outcomes

On successful completion of the course, students will be able to:

1. Understand the fundamentals of family business including its definition, importance, complexity, and how it differs from non-family business.
2. Analyse the factors contributing to the longevity of family businesses, and the causes for the potential early demise.
3. Understand the financing options of family business, and risk related to getting listed and over-borrowing.
4. Valuate the family business for different purposes, including pruning of ownership, professionalizing of family business and selling of family business.
5. Prepare for Leadership and ownership succession planning.
6. Design suitable structures and governance systems for wealth creators to preserve family wealth, legacy, and harmony across generations.

FINA 4423 Family Office 3 Credit(s)

Previous Course Code(s)	*FINA 4929P*
Prerequisite(s)	*FINA 2203 OR FINA 2303*

Description

This course provides students with a comprehensive and most up-to-date understanding of family office, as well as insiders' view and practical aspects of family office. For university students who are about to graduate and are interested in entering into family office related sectors, having an in-depth understanding of family office will definitely help them better prepared for the future.

Intended Learning Outcomes

On successful completion of the course, students will be able to:

1. Understand the fundamentals of family business and the factors contributing to the longevity of family businesses.
2. Prepare for leadership and ownership succession.
3. Understand the need for transition from family business to family office.
4. Understand the fundamentals of family office, single family office vs multi family office.
5. Understand the definition and measurements of impact investments and be able to engage it practically.
6. Understand the different approaches and impacts of family philanthropy.
7. Design a suitable family governance structure.
8. Understand art collection and investment of family office.

FINA 4503 Banking and Financial Intermediation 3 Credit(s)

Prerequisite(s) *FINA 2203 OR FINA 2303*

Description

This course examines financial intermediaries with a particular emphasis on banks. It introduces students to the nature, purpose, and management of these institutions. The course also brings together the upstream issues in risk measurement and the regulative environment surrounding banks. By the end of this course, students will be able to understand the role of banks in the economy and the effects of frictions such as information asymmetry and agency costs on bank behavior. The course will start with the review of basic game theory.

Intended Learning Outcomes

On successful completion of the course, students will be able to:

1. Explain why financial intermediaries exist in the economy and understand how they create value.
2. Explain informational frictions and agency costs in bank lending behavior.
3. Describe the risks banks take in their operations.
4. Understand the need and nature of bank regulations.

FINA 4513 Risk Management 3 Credit(s)

Prerequisite(s)	*FINA 3103 AND FINA 3203*
Cross-Campus Equivalent Course	*FTEC 4420*

Description

This course covers the role of risk management in supporting companies as they strive to balance the internal and external risk factors surrounding the operation of their business model against their various stakeholder obligations. Topics include a review of basic hedging strategies (knowledge of futures and options is pre-requisite), the theory and evidence on the value of corporate risk management, review of major surveys of risk management practices, business-case studies highlighting advanced derivatives and risk-management strategies, an illustrative model of integrated enterprise risk- management (featuring Monte-Carlo simulation), and student-led risk-management audits.

Intended Learning Outcomes

On successful completion of the course, students will be able to:

1. Understand how to identify, measure, and assess risks.
2. Design and implement advanced derivatives and hedging strategies to preserve and add value.
3. Analyze and evaluate risk factors by modeling their causes and effects through enterprise risk management frameworks.
4. Apply the risk management concepts and tools to various cases and scenarios.

FINA 4603 Venture Capital Financing 3 Credit(s)

Prerequisite(s)	*FINA 3303*

Description

This course covers the application of financial tools and techniques to the funding and valuation of an entrepreneurial venture. The course seeks to provide students with a background on the venture capital industry, the economic intuition underlying its features, and the basic tools required to work in the industry.

Intended Learning Outcomes

On successful completion of the course, students will be able to:

1. Understand the structure and key players of the venture capital industry.
2. Understand how venture capitalists identify, add, and capture value in portfolio companies.
3. Evaluate the incentives and roles of different stakeholders in the venture capital ecosystem.
4. Understand and assess the types of contracts used in VC deals, and how they influence control, risk, and returns.
5. Assess and value exit strategies, including IPOs, acquisitions, and secondary sales.

FINA 4703 ESG Investing 3 Credit(s)

Prerequisite(s) *FINA 3103*

Description

This course focuses on the relevance of sustainability factors on financial performance of firms and securities. Topics in this course include the market terminology, practices, usages and impact of environmental, social and governance (ESG) factors and climate risk. Students will learn to analyze complex financial problems, adapt investment strategies to meet business needs, propose solutions that maximize stakeholder value, and apply ESG related concepts to the process of investment management and valuation.

Intended Learning Outcomes

On successful completion of the course, students will be able to:

1. Understand the definition and importance of ESG (Environmental, Social, and Governance).
2. Understand the tradeoff between financial and social returns.
3. Describe the trends in the asset management industry and impact investing.
4. Identify the channels through which impact investing improves sustainability.
5. Explain the relationship between ESG, impact investing, and firms' performance.

FINA 4713 Introduction to Artificial Intelligence and Big Data in Finance 3 Credit(s)

Previous Course Code(s) *FINA 4929O*
Prerequisite(s) *FINA 3103*

Cross-Campus Equivalent Course	*AIAA 4330*

Description

This course provides an overview of the main branches of AI and Big Data, and exposes students to real-world industry use cases to help them understand the impact that AI will have on the finance industry. Upon completion of this course, students will be able to identify and leverage AI and Big Data to enhance and/or disrupt business models, while keeping in mind social and ethical considerations.

Intended Learning Outcomes

On successful completion of the course, students will be able to:

1. Understand the fundamental branches of Artificial Intelligence and Big Data.
2. Gain insight into the use cases of AI/Big Data within Finance and the Financial Markets.
3. Critically evaluate use cases of AI/Big Data from the perspective of data, technology, human capital and change management.
4. Identify concerns relating to the AI/Big Data from a regulatory, organizational, societal and personal standpoint.
5. Anticipate the future evolutions and potential ethical issues surrounding the impact of AI/Big Data.

– FINA 4723 Responsible Finance 3 Credit(s)

Previous Course Code(s)	*FINA 4929Q*
Prerequisite(s)	*FINA 2203 OR FINA 2303*

Description

This course is designed to give students a thorough overview of responsible finance and ESG. The first part covers fundamental topics related to ESG. The second part focuses more on project and firm valuation with an ESG perspective. It also covers financial models that will allow one to understand how to incorporate ESG factors in financial decisions.

Intended Learning Outcomes

On successful completion of the course, students will be able to:

1. Identify and explain the driving forces of responsible finance.
2. Understand the objective of corporation.
3. Understand how investors can engage with firms in effective manners.
4. Apply tools and data sets to evaluate ESG performance.
5. Analyze corporate projects using ESG criteria.
6. Evaluate firm performance using ESG methods.

FINA 4803 Quantitative Trading 3 Credit(s)

Exclusion(s)	*FINA 4810, FINA 4814*

Description

For QFIN students in their fourth year of study only. This is a capstone course for final year students in the BSc Quantitative Finance program. Working in groups, the students will bring together all the knowledge learned in the program to develop and execute a quantitative trading strategy through a simulated platform. The instructor will guide the students through the following stages: (1) idea generation, (2) strategy backtesting, (3) risk management planning, (4) system development, (5) trade execution, and (6) performance evaluation. Students will also have the opportunity to develop their leadership and presentation skills.

Intended Learning Outcomes

On successful completion of the course, students will be able to:

1. Develop a quantitative trading strategy.
2. Develop a risk management overlay.
3. Execute a trading strategy effectively.
4. Evaluate a trading strategy's performance.
5. Present a trading strategy in a professional manner.

FINA 4919 Independent Study 1-4 Credit(s)

Description

Directed study of selected issues in Finance. May be graded by letter, or P/F for different offerings. Instructor's approval is required for enrollment in the course. Subject to approval by the department, students may repeat the course once if a different topic is selected.

Intended Learning Outcomes

On successful completion of the course, students will be able to:

1. Demonstrate an understanding of finance topics through independent research and literature review.
2. Apply appropriate data collection and analysis techniques to investigate a finance-related research question.
3. Generate insights and support evidence-based conclusions.
4. Communicate research findings clearly and professionally through written reports and/or oral presentations.

FINA 4929 Special Topics in Finance 0-4 Credit(s)

Description

Covers current developments in finance. Topics selected by instructors. May be graded P or F.

Intended Learning Outcomes

On successful completion of the course, students will be able to:

1. Demonstrate an understanding of key concepts, frameworks, or practices relevant to the course topic.
2. Analyze issues related to the course content using appropriate reasoning, frameworks, or tools.
3. Interpret and evaluate case studies, scenarios, or other relevant materials using critical thinking and appropriate analytical reasoning.

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